

Optimal energy decay rate for a kinetic equation coupled with fluid equations

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Backgrounds

When a suspension of active particles is in a fluid, the dynamics of the system can be modeled by a kinetic equation coupled with fluid equations. The position $\mathbf{x}_i(t) \in \mathbb{R}^n$ and velocity $\mathbf{v}_i(t) \in \mathbb{R}^n$ of the i -th particle are governed by the following system:

$$\begin{cases} \frac{d\mathbf{x}_i}{dt} = \mathbf{u} + \mathbf{v}_i, \\ \frac{d\mathbf{v}_i}{dt} = \frac{1}{N} \sum_{j=1}^N \psi(|\mathbf{x}_j - \mathbf{x}_i|)(\mathbf{v}_j - \mathbf{v}_i) + \mathbf{u} - \mathbf{v}_i, \quad i = 1, 2, \dots, N, \end{cases} \quad (1.1)$$

where $\mathbf{u}$ is the fluid velocity, and $\varphi(\cdot)$ represents the interaction function between particles.

- If $|\mathbf{v}_i| \gg |\mathbf{u}|$, then $\mathbf{u} + \mathbf{v}_i \approx \mathbf{v}_i$. Formally taking a mean-field limit leads to the following kinetic equation

$$f_t + \nabla_x \cdot (vf) + \nabla_v \cdot (L[f]f + (\mathbf{u} - v)f) = 0.$$

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$$f_t + \nabla_x \cdot (\mathbf{u}f) + \nabla_v \cdot (L[f]f + (\mathbf{u} - v)f) = 0.$$

Here, $\mathbf{u}$ is the fluid velocity, $f(t, x, v)$ is the particle distribution function in phase space $(x, v) \in \mathbb{R}^n \times \mathbb{R}^n$ at time t , and $L[f]$ is given by

$$L[f](t, x, v) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \varphi(|x - y|) f(t, y, v^*) (v^* - v) dy dv^*.$$

The kinetic-fluid models were ever introduced in the combustion theory by Williams in 1985. Such models have received increasing attention in recent years. Most previous research focused on the following two type of results:

- Weak or strong solutions in a spatial-periodic or bounded domain;
- Classical solutions around an equilibrium.

Previous research

Weak or strong solutions in a spatial-periodic or bounded domain

- Hamdache (Japan J. Indust. Appl. Math.,1998): Vlasov+Stokes
Global existence and large time behaviors of weak solutions in a bounded domain with a reflection condition;
- Boudin-Desvillettes-Grandmont-Moussa (Diff. Integral Eqns., 2009):
Vlasov+INS
Global existence of weak solutions in a spatial-periodic domain;
- Bae-Choi-Ha-Kang: Kinetic CS+Fluid equations
 - 1) Kinetic CS + INS(Nonlinearity,2012): Global existence of weak solutions in a spatial-periodic domain+Large time behaviors
 - 2) Kinetic CS + INS(JDE,2014): Local existence of strong solutions in a spatial-periodic domain+Large time behaviors
 - 3) Kinetic CS + CNS(DCDS,2014): Local existence of strong solutions in a spatial-periodic domain+Large time behaviors
 - 4) Kinetic CS + Stokes(JMFM,2016): Global existence of strong solutions in a spatial-periodic domain

Classical solutions around an equilibrium

- Carrillo-Duan-Moussa (KRM,2011): VFP+Euler
Global existence of classical solutions+large time behaviors;
- Duan-Liu (KRM,2013): VFP+Compressible Euler
Global existence of classical solutions+large time behaviors;
- Goudon-He-Moussa-Zhang (SIAM,2010): VFP+NS
Global existence of classical solutions in a spatial-periodic domain+large time behaviors+regularity;
- Li-Mu-Wang (SIAM,2017): VFP+Compressible NS
Global existence of classical solutions+large time behaviors

However, the following two problems seems to be less investigated for such kinetic-fluid systems.

- Under what conditions on initial data, the Cauchy problem of a kinetic-fluid system admits a global-in-time strong solution.
- If a initial data is not around an equilibrium, what is the large time behavior of a kinetic-fluid system?

As far as we know, recently Han-Kwan (Probab. Math. Phys., 2022) studied the Cauchy problem of a Vlasov–Navier–Stokes equations.

Under some smallness assumption on initial data, he obtained global existence of weak solutions, and showed that the energy of the system decays at the rate $(1+t)^{-\alpha}$ for $\alpha \in (0, \frac{3}{2})$.

However, the energy decay rate obtained by Han-Kwan is **suboptimal**.

Recently, we develop a new method to deal with the coupling term, which can improve Han-Kwan's result and obtain optimal energy decay rate. In order to simply demonstrate our main idea, we focus on the following system:

$$\begin{cases} f_t + \mathbf{u} \cdot \nabla_x f + \nabla_v \cdot ((\mathbf{u} - v)f) = 0, \\ \mathbf{u}_t + \nabla P = \Delta \mathbf{u} + \int_{\mathbb{R}^n} (v - \mathbf{u}) f dv, \\ \nabla \cdot \mathbf{u} = 0, \end{cases} \quad (2.1)$$

subject to the initial data

$$f|_{t=0} = f_0, \quad \mathbf{u}|_{t=0} = \mathbf{u}_0, \quad (2.2)$$

with $\mathbf{u}_0$ satisfying the compatibility condition $\nabla \cdot \mathbf{u}_0 = 0$.

Our main aim is to deal with the coupling term. We choose the transport term $\mathbf{u} \cdot \nabla_x f$ instead of $v \cdot \nabla_x f$, in order to avoid the difficulty in the estimate of uniform-in-time bound on the particle density.

Remark

Such transport term as in the first kinetic equation also appeared in some micro-macro models from polymeric fluids, cf. Constantin–Masmoudi (CMP, 2007), Lin–Liu–Zhang (CPAM, 2007), and Lin–Zhang–Zhang (CMP, 2008).

Since the kinetic equation is coupled with Stokes equations, global classical solution is easy to obtain for some regular initial data. Our interest lies in energy decay rate of the coupled system.

Define the particle density as

$$\rho(t, x) := \int_{\mathbb{R}^n} f(t, x, v) dv,$$

with $\rho_0(x) := \rho(0, x)$. The energy of the system (2.1) is defined as

$$E(t) := \frac{1}{2} \|\mathbf{u}(t)\|_{L^2}^2 + \frac{1}{2} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(t, x, v) |v|^2 dx dv,$$

with $E_0 := E(0)$. The v -support of $f(t, x, v)$ at time t is defined as

$$\text{supp}_v f := \left\{ v : (x, v) \in \text{supp} f(t, \cdot, \cdot) \right\}.$$

Denote by $B(R_0)$ the ball centered at the origin with a radius $R_0 \in (0, \infty)$.

We have the following theorem.

Jin(AML, 2024)

Let $n \geq 3$. Assume that $f_0(x, v) \geq 0$, $f_0(x, v) \in L^1(\mathbb{R}^n \times \mathbb{R}^n) \cap L^\infty(\mathbb{R}^n \times \mathbb{R}^n)$, $\text{supp}_v f_0 \subseteq B(R_0)$, and $\mathbf{u}_0(x) \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$ with $\nabla \cdot \mathbf{u}_0 = 0$. For any classical solution to the system (2.1)–(3.2), the energy $E(t)$ decays at least at the rate $(1+t)^{-\frac{n}{2}}$. To be precise,

$$E(t) \leq C(1+t)^{-\frac{n}{2}} \quad \forall t \in [0, \infty),$$

where C is some constant depending only on E_0 , $\|\rho_0\|_{L^\infty}$, $\|f_0\|_{L^1}$ and $\|\mathbf{u}_0\|_{L^1}$.

Remark

The energy decay rate is optimal in the sense that the energy of the system decays at a rate coinciding with the one corresponding to the underlying linear parabolic equations.

Define

$$\mathbf{h}(t, x) := \int_{\mathbb{R}^n} (v - \mathbf{u}) f dv.$$

Then we show how to deal with the coupling term to obtain the optimal energy decay rate.

Outline of the proof

Step 1. Uniform-in-time bound on $\|\rho(t)\|_{L^\infty}$

Integrating (2.1)₁ with respect to v over $\mathbb{R}^3$ gives

$$\rho_t + \mathbf{u} \cdot \nabla \rho = 0. \quad (2.3)$$

Multiplying (2.3) by $p\rho^{p-1}$, $1 \leq p < \infty$, and integrating the resulting equation over $[0, t] \times \mathbb{R}^3$, we obtain that for $t \in [0, T]$,

$$\|\rho(t)\|_{L^p} = \|\rho_0\|_{L^p} \leq \|\rho_0\|_{L^1}^{\frac{1}{p}} \|\rho_0\|_{L^\infty}^{1-\frac{1}{p}} \leq C \|f_0\|_{L^\infty}^{1-\frac{1}{p}} R_0^{3-\frac{3}{p}} \quad (2.4)$$

for some constant $C > 0$, where we have used the incompressible condition $\nabla \cdot \mathbf{u} = 0$. Then,

$$\|\rho(t)\|_{L^\infty} = \lim_{p \rightarrow \infty} \|\rho(t)\|_{L^p} = \lim_{p \rightarrow \infty} \|\rho_0\|_{L^p} = \|\rho_0\|_{L^\infty} \leq C \|f_0\|_{L^\infty} R_0^3.$$

Step 2. Derive a Gronwall inequality for $E(t)$

An elementary estimate gives

$$\frac{d}{dt}E(t) + \|\nabla \mathbf{u}\|_{L^2}^2 + \int_{\mathbb{R}^{2n}} |v - \mathbf{u}|^2 f dx dv \leq 0. \quad (2.5)$$

Denote the Fourier transform of $\mathbf{u}(t, x)$ in x by

$$\hat{\mathbf{u}}(t, \xi) := \int_{\mathbb{R}^n} \mathbf{u}(t, x) e^{-2\pi i \xi \cdot x} dx.$$

It is easy to see that

$$\begin{aligned}\|\nabla \mathbf{u}\|_{L^2}^2 &= \int_{\mathbb{R}^n} 4\pi^2 |\hat{\mathbf{u}}(t, \xi)|^2 |\xi|^2 d\xi \\ &\geq \int_{|\xi| > \frac{c}{2\pi} (t+c^2)^{-\frac{1}{2}}} 4\pi^2 |\hat{\mathbf{u}}(t, \xi)|^2 |\xi|^2 d\xi \\ &\geq \frac{c^2}{t+c^2} \int_{|\xi| > \frac{c}{2\pi} (t+c^2)^{-\frac{1}{2}}} |\hat{\mathbf{u}}(t, \xi)|^2 d\xi \\ &= \frac{c^2}{t+c^2} \int_{\mathbb{R}^n} |\mathbf{u}(t, x)|^2 dx - \frac{c^2}{t+c^2} \int_{|\xi| \leq \frac{c}{2\pi} (t+c^2)^{-\frac{1}{2}}} |\hat{\mathbf{u}}(t, \xi)|^2 d\xi,\end{aligned}\tag{2.6}$$

for some constant $c > 0$ to be determined later, where we have used the Plancherel theorem to show that

$$\|\hat{\mathbf{u}}(t)\|_{L^2}^2 = \|\mathbf{u}(t)\|_{L^2}^2.$$

Substituting (2.6) into (2.5), we deduce for $t \geq 0$ that

$$\begin{aligned} \frac{d}{dt}E(t) + \frac{c^2}{t+c^2} \left(\|\mathbf{u}\|_{L^2}^2 + \int_{\mathbb{R}^{2n}} |\mathbf{v} - \mathbf{u}|^2 f dx dv \right) \\ \leq \frac{c^2}{t+c^2} \int_{|\xi| \leq \frac{c}{2\pi}} |\hat{\mathbf{u}}(t, \xi)|^2 d\xi. \end{aligned} \quad (2.7)$$

Note that

$$\begin{aligned} & \|\mathbf{u}\|_{L^2}^2 + \int_{\mathbb{R}^{2n}} |\mathbf{v}|^2 f dx dv \\ & \leq \|\mathbf{u}\|_{L^2}^2 + 2 \int_{\mathbb{R}^{2n}} |\mathbf{v} - \mathbf{u}|^2 f dx dv + 2 \int_{\mathbb{R}^{2n}} |\mathbf{u}|^2 f dx dv \\ & \leq \|\mathbf{u}\|_{L^2}^2 + 2 \int_{\mathbb{R}^{2n}} |\mathbf{v} - \mathbf{u}|^2 f dx dv + 2 \|\rho_0\|_{L^\infty} \|\mathbf{u}\|_{L^2}^2 \\ & \leq 2(1 + \|\rho_0\|_{L^\infty}) \left(\|\mathbf{u}\|_{L^2}^2 + \int_{\mathbb{R}^{2n}} |\mathbf{v} - \mathbf{u}|^2 f dx dv \right), \end{aligned} \quad (2.8)$$

Substituting (2.8) into (2.7) gives

$$\frac{d}{dt}E(t) + \frac{c^2}{(t+c^2)(1+\|\rho_0\|_{L^\infty})}E(t) \leq \frac{c^2}{t+c^2} \int_{|\xi| \leq \frac{c}{2\pi}} (t+c^2)^{-\frac{1}{2}} |\hat{\mathbf{u}}(t, \xi)|^2 d\xi. \quad (2.9)$$

Take $c^2 = n(1 + \|\rho_0\|_{L^\infty})$. Then (2.9) becomes

$$\frac{d}{dt}E(t) + \frac{n}{t+c^2}E(t) \leq \frac{c^2}{t+c^2} \int_{|\xi| \leq \frac{c}{2\pi}} (t+c^2)^{-\frac{1}{2}} |\hat{\mathbf{u}}(t, \xi)|^2 d\xi. \quad (2.10)$$

Step 3. Estimate on $\hat{\mathbf{u}}(t, \xi)$ and the proof by contradiction

Applying the Fourier transform to $(2.1)_2$ gives

$$\hat{\mathbf{u}}_t + \widehat{\nabla P} = -4\pi^2 |\xi|^2 \hat{\mathbf{u}} + \hat{\mathbf{h}}. \quad (2.11)$$

It follows from (2.11) that

$$|\hat{\mathbf{u}}(t, \xi)| \leq |\hat{\mathbf{u}}_0(\xi)| e^{-4\pi^2 |\xi|^2 t} + \int_0^t e^{-4\pi^2 |\xi|^2 (t-s)} \left(|\widehat{\nabla P}|(s, \xi) + |\hat{\mathbf{h}}|(s, \xi) \right) ds. \quad (2.12)$$

Taking divergence to $(2.1)_2$ yields

$$\Delta P = \nabla \cdot \mathbf{h}. \quad (2.13)$$

From (2.13), we deduce that

$$|\widehat{\nabla P}|(s, \xi) \leq C |\hat{\mathbf{h}}|(s, \xi), \quad (2.14)$$

for some constant $C > 0$.

Substituting (2.14) into (2.12), we obtain

$$\begin{aligned} |\hat{\mathbf{u}}(t, \xi)| &\leq |\hat{\mathbf{u}}_0(\xi)| e^{-4\pi^2 |\xi|^2 t} + C \int_0^t e^{-4\pi^2 |\xi|^2 (t-s)} |\hat{\mathbf{h}}(s, \xi)| ds \\ &\leq \|\mathbf{u}_0\|_{L^1} + C \int_0^t \|\mathbf{h}(s)\|_{L^1} ds, \end{aligned} \tag{2.15}$$

for some constant $C > 0$.

Remark

The difficulty lies in estimate on the coupling term $\int_0^t \|\mathbf{h}(s)\|_{L^1} ds$.

An elementary energy estimate gives that

$$E(t) \leq E_0 \quad \text{for all } t \in [0, \infty).$$

Thus, there must exist $T > 0$ and some constant $K > 1$, such that

$$E(t) < K(1+t)^{-\frac{n}{2}+\frac{1}{8}} \quad \text{for } t \in [0, T]. \quad (2.16)$$

Denote by T_m the supremum of all the T in (2.16). Next we prove by contradiction that $T_m = \infty$.

Suppose not, i.e., $T_m < \infty$. Then it holds that

$$E(T_m) = K(1 + T_m)^{-\frac{n}{2} + \frac{1}{8}}. \quad (2.17)$$

Multiplying (2.5) by $(1 + t)^{\frac{9}{8}}$ gives

$$\frac{d}{dt} \left((1 + t)^{\frac{9}{8}} E(t) \right) + (1 + t)^{\frac{9}{8}} \int_{\mathbb{R}^{2n}} |v - u|^2 f dx dv \leq \frac{9}{8} (1 + t)^{\frac{1}{8}} E(t). \quad (2.18)$$

We integrate (2.18) over $[0, T_m]$ to obtain

$$\int_0^{T_m} (1 + t)^{\frac{9}{8}} \int_{\mathbb{R}^{2n}} |v - u|^2 f dx dv dt \leq E_0 + CK \quad (2.19)$$

for some constant $C > 0$.

From (2.19), we infer that

$$\begin{aligned}
 C \int_0^t \|\mathbf{h}(s)\|_{L^1} ds &\leq C \int_0^{T_m} (1+s)^{-\frac{9}{16}} (1+s)^{\frac{9}{16}} \|\mathbf{h}(s)\|_{L^1} ds \\
 &\leq C \left(\int_0^{T_m} (1+s)^{\frac{9}{8}} \int_{\mathbb{R}^{2n}} |v - \mathbf{u}|^2 f dx dv ds \right)^{\frac{1}{2}} \quad (2.20) \\
 &\leq C(E_0, \|f_0\|_{L^1}) (1+K)^{\frac{1}{2}}.
 \end{aligned}$$

Substituting (2.20) into (2.15), we deduce that

$$|\hat{\mathbf{u}}(t, \xi)|^2 \leq C + CK \quad \text{for } t \in [0, T_m], \quad (2.21)$$

where $C := C(\|\mathbf{u}_0\|_{L^1}, \|f_0\|_{L^1}, E_0)$.

We deduce from (2.10) and (2.21) that

$$E(t) \leq CK(1+t)^{-\frac{n}{2}} \quad \text{for } t \in [0, T_m], \quad (2.22)$$

where $C := C(\|\rho_0\|_{L^\infty}, \|\mathbf{u}_0\|_{L^1}, \|f_0\|_{L^1}, E_0)$. From

$$E(T_m) = K(1+T_m)^{-\frac{n}{2}+\frac{1}{8}} \leq E_0,$$

we have

$$1+T_m \geq \left(\frac{K}{E_0}\right)^{\frac{8}{4n-1}}. \quad (2.23)$$

From (2.22) and (2.23), we infer that

$$\begin{aligned} E(T_m) &\leq C(1 + T_m)^{-\frac{1}{8}} K(1 + T_m)^{-\frac{n}{2} + \frac{1}{8}} \\ &\leq CK^{-\frac{1}{4n-1}} K(1 + T_m)^{-\frac{n}{2} + \frac{1}{8}}, \end{aligned} \quad (2.24)$$

for some constant $C := C(\|\rho_0\|_{L^\infty}, \|\mathbf{u}_0\|_{L^1}, \|f_0\|_{L^1}, E_0)$. Take K suitably large such that

$$CK^{-\frac{1}{4n-1}} = \frac{1}{2}.$$

Then we have $E(T_m) \leq \frac{K}{2}(1 + T_m)^{-\frac{n}{2} + \frac{1}{8}}$, which contradicts (2.17). Therefore, we conclude $T_m = \infty$ by contradiction, and it follows from (2.22) that

$$E(t) \leq C(1 + t)^{-\frac{n}{2}} \quad (2.25)$$

for all $t \geq 0$ and some constant $C := C(\|\rho_0\|_{L^\infty}, \|\mathbf{u}_0\|_{L^1}, \|f_0\|_{L^1}, E_0)$. $\square$

Another kinetic-fluid model

A kinetic equation coupled with the incompressible Navier–Stokes equations

$$\begin{cases} f_t + \mathbf{u} \cdot \nabla_x f + \nabla_v \cdot (L[f]f + (\mathbf{u} - v)f) = 0, \\ \mathbf{u}_t + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla P = \Delta \mathbf{u} + \int_{\mathbb{R}^3} (v - \mathbf{u}) f dv, \\ \nabla \cdot \mathbf{u} = 0, \end{cases} \quad (3.1)$$

subject to the initial data

$$f|_{t=0} = f_0, \quad \mathbf{u}|_{t=0} = \mathbf{u}_0, \quad (3.2)$$

with $\mathbf{u}_0$ satisfying the compatibility condition $\nabla \cdot \mathbf{u}_0 = 0$.

As for the above system, we also study the following two problems:

- Global existence of strong solutions to the Cauchy problem;
- Energy decay rate of the coupled system.

We introduce the following **weighted Sobolev space** to overcome the difficulty caused by unboundedness of the domain.

$$H_{\omega}^1(\mathbb{R}^3 \times \mathbb{R}^3) := \left\{ h(x, v) : h \in L_{\omega}^2(\mathbb{R}^3 \times \mathbb{R}^3), \right. \\ \left. \nabla_x h \in L_{\omega}^2(\mathbb{R}^3 \times \mathbb{R}^3), \nabla_v h \in L_{\omega}^2(\mathbb{R}^3 \times \mathbb{R}^3) \right\},$$

$$\|h\|_{H_{\omega}^1}^2 := \|h\|_{L_{\omega}^2}^2 + \|\nabla_x h\|_{L_{\omega}^2}^2 + \|\nabla_v h\|_{L_{\omega}^2}^2,$$

where

$$\|h\|_{L_{\omega}^2} := \left(\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} h^2(x, v) \omega(x) dx dv \right)^{\frac{1}{2}}, \quad \omega(x) := (1 + |x|^2)^{\alpha}, \quad \alpha > \frac{3}{2},$$

similarly for $\|\nabla_x h\|_{L_{\omega}^2}$ and $\|\nabla_v h\|_{L_{\omega}^2}$.

Our theorem

Jin (submitted)

Let $0 < R_0 < \infty$. Assume that $f_0(x, v) \geq 0$, $f_0(x, v) \in H^1_\omega(\mathbb{R}^3 \times \mathbb{R}^3) \cap L^\infty(\mathbb{R}^3 \times \mathbb{R}^3)$, and $\mathbf{u}_0(x) \in H^1(\mathbb{R}^3)$, with $\nabla \cdot \mathbf{u}_0 = 0$ and the v -support of $f_0(x, v)$ satisfying

$$\text{supp}_v f_0(x, \cdot) \subseteq B(R_0) \quad \text{for a.e. } x \in \mathbb{R}^3.$$

If there is a sufficiently small constant ε_0 such that

$$C \|\mathbf{u}_0\|_{L^3}^6 + (1 + \|\rho_0\|_{L^\infty}) E_0 \leq \varepsilon_0$$

for some constant $C > 0$ independent of the initial data, then the Cauchy problem (3.1)–(3.2) admits a unique global-in-time strong solution. Moreover, if $\mathbf{u}_0(x) \in L^1(\mathbb{R}^3)$ in addition, then

$$E(t) \leq C(1+t)^{-\frac{3}{2}} \quad \forall t \in [0, \infty),$$

where $C := C(E_0, \|\rho_0\|_{L^\infty}, \|\mathbf{u}_0\|_{L^1})$.

Remark

The smallness assumption on initial data is used to prove existence of global-in-time strong solution to the coupled system.

The proof of existence of global strong solutions consists of the following two steps:

- (1) Construction of local strong solution by iteration;
- (2) Some a priori estimates used to extend the local strong solution to all the time.

The first step is standard. In the second step, the difficulty is to deal with the convection term $\mathbf{u} \cdot \nabla \mathbf{u}$.

We need to estimate $\sup_{0 \leq t \leq T} \|\nabla \mathbf{u}(t)\|_{L^2}$ for all $T > 0$. However, for incompressible Navier–Stokes equations, this estimate can only be obtained under some smallness assumption on initial data so far.

Estimate on $\nabla \mathbf{u}$

Taking the inner product of (3.1)₂ with $-\Delta \mathbf{u}$, we obtain

$$\begin{aligned}
 & \frac{d}{dt} \|\nabla \mathbf{u}\|_{L^2}^2 + \lambda_1 \|\nabla^2 \mathbf{u}\|_{L^2}^2 \\
 & \leq \int_{\mathbb{R}^3} \mathbf{u} \cdot \nabla \mathbf{u} \cdot \Delta \mathbf{u} dx - \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} (\nu - \mathbf{u}) f d\nu \cdot \Delta \mathbf{u} dx \\
 & \leq \|\mathbf{u}\|_{L^9} \|\nabla \mathbf{u}\|_{L^{\frac{18}{7}}} \|\Delta \mathbf{u}\|_{L^2} + \|\mathbf{h}\|_{L^2} \|\Delta \mathbf{u}\|_{L^2} \\
 & \leq \frac{\lambda_1}{2} \|\nabla^2 \mathbf{u}\|_{L^2}^2 + C \|\mathbf{u}\|_{L^9}^3 \|\nabla \mathbf{u}\|_{L^2}^2 + C \|\mathbf{h}\|_{L^2}^2,
 \end{aligned} \tag{3.3}$$

for some positive constants λ_1 and C , where we have used the elliptic estimate to deduce that

$$\lambda_1 \|\nabla^2 \mathbf{u}\|_{L^2}^2 \leq \|\Delta \mathbf{u}\|_{L^2}^2.$$

Estimate on $\|\mathbf{u}\|_{L^3(0,T;L^9)}$

Using the splitting approach, we decompose $\mathbf{u} = \bar{\mathbf{u}} + \mathbf{w}$, with $\bar{\mathbf{u}}$ and $\mathbf{w}$ determined by the following equations:

$$\begin{cases} \bar{\mathbf{u}}_t + \mathbf{u} \cdot \nabla \bar{\mathbf{u}} = \Delta \bar{\mathbf{u}}, \\ \bar{\mathbf{u}}|_{t=0} = \mathbf{u}_0, \end{cases} \quad (3.4)$$

and

$$\begin{cases} \mathbf{w}_t + \mathbf{u} \cdot \nabla \mathbf{w} + \nabla P = \Delta \mathbf{w} + \int_{\mathbb{R}^3} (\nu - \mathbf{u}) f d\nu, \\ \mathbf{w}|_{t=0} = 0, \end{cases} \quad (3.5)$$

respectively.

L^3 -type energy estimate

Taking the dot product of (3.4) with $3|\bar{\mathbf{u}}|\bar{\mathbf{u}}$, and integrating the resulting equation over $\mathbb{R}^3$, we obtain

$$\frac{d}{dt} \|\bar{\mathbf{u}}\|_{L^3}^3 + 3 \int_{\mathbb{R}^3} |\bar{\mathbf{u}}| |\nabla \bar{\mathbf{u}}|^2 dx \leq 0. \quad (3.6)$$

Taking the dot product of (3.5) with $3|\mathbf{w}|\mathbf{w}$, and integrating the resulting equation over $\mathbb{R}^3$, we deduce that

$$\frac{d}{dt} \|\mathbf{w}\|_{L^3}^3 + 3 \int_{\mathbb{R}^3} |\mathbf{w}| |\nabla \mathbf{w}|^2 dx \leq -3 \int_{\mathbb{R}^3} |\mathbf{w}| \mathbf{w} \cdot \nabla P dx + 3 \int_{\mathbb{R}^3} |\mathbf{w}| \mathbf{w} \cdot \mathbf{h} dx, \quad (3.7)$$

where $\mathbf{h}(t, x)$ is defined as

$$\mathbf{h}(t, x) := \int_{\mathbb{R}^3} (v - \mathbf{u}) f dv.$$

In the process of the L^3 -type energy estimate, the difficulty mainly arises from the term $\int_{\mathbb{R}^3} 3|\mathbf{w}|\mathbf{w} \cdot \nabla P dx$, where the pressure P is determined by the following equation:

$$\Delta P = -\nabla \cdot (\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla \cdot \mathbf{h}, \quad \text{with } \mathbf{h} := \int_{\mathbb{R}^3} (\nu - \mathbf{u}) f dv.$$

We overcome this difficulty by decomposing $P = P_1 + P_2$, with P_1 and P_2 governed by

$$\Delta P_1 = -\nabla \cdot (\mathbf{u} \cdot \nabla \mathbf{u}) \quad \text{and} \quad \Delta P_2 = \nabla \cdot \mathbf{h},$$

respectively.

Then it follows from the Calderón–Zygmund theorem that

$$\|P_1\|_{L^{\frac{9}{4}}} \leq C\|\mathbf{u}\|_{L^{\frac{9}{2}}}^2 \quad \text{and} \quad \|\nabla P_2\|_{L^{\frac{3}{2}}} \leq C\|\mathbf{h}\|_{L^{\frac{3}{2}}},$$

for some constant $C > 0$. We estimate $3 \int_{\mathbb{R}^3} 3|\mathbf{w}|\mathbf{w} \cdot \nabla P dx$ as follows:

$$\begin{aligned} & 3 \int_{\mathbb{R}^3} |\mathbf{w}|\mathbf{w} \cdot \nabla P dx \\ & \leq 3 \int_{\mathbb{R}^3} |\mathbf{w}|\mathbf{w} \cdot (\nabla P_1 + \nabla P_2) dx \\ & \leq -3 \int_{\mathbb{R}^3} \nabla \cdot (|\mathbf{w}|\mathbf{w}) P_1 dx + 3 \int_{\mathbb{R}^3} |\mathbf{w}|\mathbf{w} \cdot \nabla P_2 dx \\ & \leq C \int_{\mathbb{R}^3} |\mathbf{w}| \cdot |\nabla \mathbf{w}| \cdot |P_1| dx + C\|\mathbf{w}\|_{L^6}^2 \|\nabla P_2\|_{L^{\frac{3}{2}}} \\ & \leq C \left\| |\mathbf{w}|^{\frac{1}{2}} |\nabla \mathbf{w}| \right\|_{L^2} \|\mathbf{w}\|_{L^9}^{\frac{1}{2}} \|P_1\|_{L^{\frac{9}{4}}} + C\|\mathbf{w}\|_{L^6}^2 \|\mathbf{h}\|_{L^{\frac{3}{2}}} \\ & \leq C \left\| |\mathbf{w}|^{\frac{1}{2}} |\nabla \mathbf{w}| \right\|_{L^2} \|\mathbf{w}\|_{L^9}^{\frac{1}{2}} (\|\bar{\mathbf{u}}\|_{L^{\frac{9}{2}}}^2 + \|\mathbf{w}\|_{L^{\frac{9}{2}}}^2) + C\|\mathbf{w}\|_{L^6}^2 \|\mathbf{h}\|_{L^{\frac{3}{2}}}. \end{aligned} \tag{3.8}$$

Define

$$I(\bar{\mathbf{u}}) := \int_{\mathbb{R}^3} |\bar{\mathbf{u}}| |\nabla \bar{\mathbf{u}}|^2 dx \quad \text{and} \quad I(\mathbf{w}) := \int_{\mathbb{R}^3} |\mathbf{w}| |\nabla \mathbf{w}|^2 dx.$$

Integrating (3.6) over $[0, T]$ gives

$$\sup_{0 \leq t \leq T} \|\bar{\mathbf{u}}(t)\|_{L^3}^3 + 3 \int_0^T I(\bar{\mathbf{u}}) dt \leq \|\mathbf{u}_0\|_{L^3}^3. \quad (3.9)$$

Under some smallness condition on initial data, we deduce

$$\sup_{0 \leq t \leq T} \|\mathbf{w}(t)\|_{L^3}^3 + \int_0^T I(\mathbf{w}) dt \leq C \|\mathbf{u}_0\|_{L^3}^6 + (1 + \|\rho_0\|_{L^\infty}) E_0. \quad (3.10)$$

By the Sobolev inequality, we have

$$\|\bar{\mathbf{u}}\|_{L^9}^3 \leq C I(\bar{\mathbf{u}}) \quad \text{and} \quad \|\mathbf{w}\|_{L^9}^3 \leq C I(\mathbf{w}) \quad (3.11)$$

for some constant $C > 0$. Then

$$\begin{aligned} \|\mathbf{u}\|_{L^3(0,T;L^9)}^3 &\leq C \left(\|\bar{\mathbf{u}}\|_{L^3(0,T;L^9)}^3 + \|\mathbf{w}\|_{L^3(0,T;L^9)}^3 \right) \\ &\leq C \int_0^T I(\bar{\mathbf{u}}) dt + C \int_0^T I(\mathbf{w}) dt \\ &\leq C \left(\|\mathbf{u}_0\|_{L^3}^3 + \|\mathbf{u}_0\|_{L^3}^6 + (1 + \|\rho_0\|_{L^\infty}) E_0 \right), \end{aligned} \quad (3.12)$$

for some constant $C > 0$ independent of the initial data. The energy decay rate of the coupled system can be obtained by the technique in Main result I. □

Problem

Some problem to be investigated

- Whether our theorem holds for large weak solution to the coupled system.
- If the transport term is $v \cdot \nabla f$ in the kinetic equation, whether our theorem still holds or not. In this case, it is difficult to derive a uniform-in-time bound on particle density.

Thanks for Your Attention!